Thin Layer Chromatography

Column Chromatography

CM3191

Dr. Zhang Sheng

Chromatography

Chromatography is a technique which the components of a mixture is separated based on differences in the rates at which they are carried through a fixed or stationary phase by a gaseous or liquid mobile phase.

The stationary phase in chromatography is a phase that is fixed in place either in a column or on a planar surface.

The mobile phase in chromatography is a phase that moves over or through the stationary phase carrying with it the analyte mixture. The mobile phase may be a gas, a liquid, or a supercritical fluid.

Technique	Stationary phase	Mobile phase
Paper chromatography	Solid (cellulose)	Liquid
Thin layer (planar) chromatography	Solid (silica or alumina)	Liquid
Column chromatography	Solid (silica or alumina)	Liquid

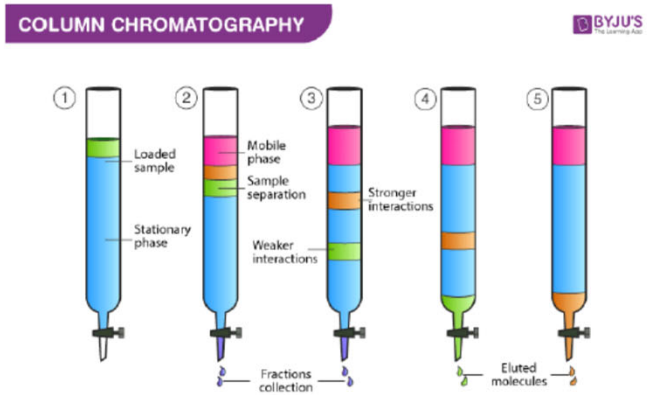
from CM2143 lecture notes

Classification of chromatographic methods

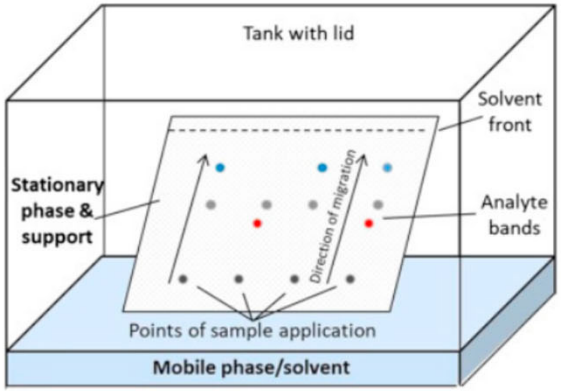
Two basic types of chromatography: Planar and column chromatography.

In *column chromatography,* the stationary phase is held in a narrow tube, and the mobile phase is forced through the tube under pressure or by gravity. In *planar chromatography,* the stationary phase is supported on a flat plate or in the pores of a paper.

In planar chromatography, the stationary phase is supported on a flat plate or in the pores of a paper and the mobile phase moves through the stationary phase by capillary action or under the influence of gravity.



https://byjus.com/chemistry/column-chromatography/



https://www.sciencedirect.com/topics/materials-science/planar-chromatography

from CM2143 lecture notes

Chromatography: General

- Thin Layer Chromatography
- Column Chromatography
- Other chromatography: GC, HPLC, GPC, IC, …

- Partition Chromatography: stationary phase is a thin liquid film adsorbed on the surface of an inert support
- Adsorption Chromatography: stationary phase is a finely-divided solid adsorbent
  - TLC
  - Column Chromatography

Chromatography: Application

- Analytical tools to establish the components of a mixture, check the purity of samples
  - Establish the purity and authenticity of starting materials
  - Monitor the reaction progress
  - Check the isolation and purification procedures

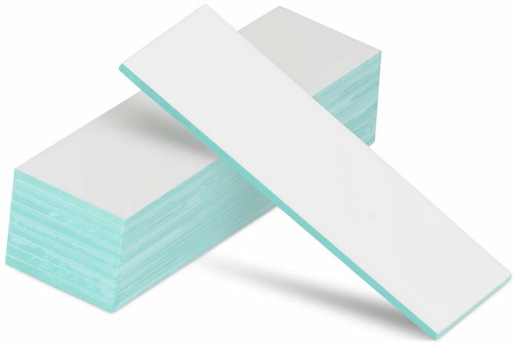
- Preparative tools for the separation of mixtures into individual components
  - Achieve the separation of product mixtures which are unable to be separated by means of distillation, recrystallization or sublimation

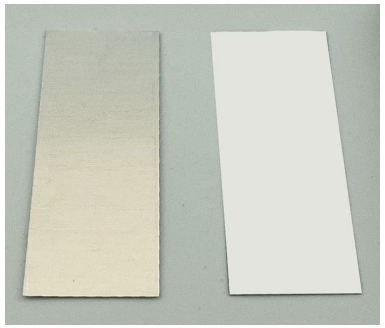
Part 1

Thin Layer Chromatography

Thin-Layer Chromatography: Chromaplate

- A thin layer of silica gel (most often used) or other finely-divided particles (e.g. alumina, celite, cellulose), coated onto the surface of a plate (glass, aluminium foil, solvent-resistant polyester sheet), by virtue of a binding agent (calcium sulphate)
- The plate is called a "chromaplate"
- Typically, a fluorescent compound (e.g. ZnS) is incorporated, to facilitate the detection of the resolved components under UV light
- For most separations, commercial chromaplate is sufficient, although the chromaplate can be prepared in the lab, particularly when the special absorbent is not commercially available.





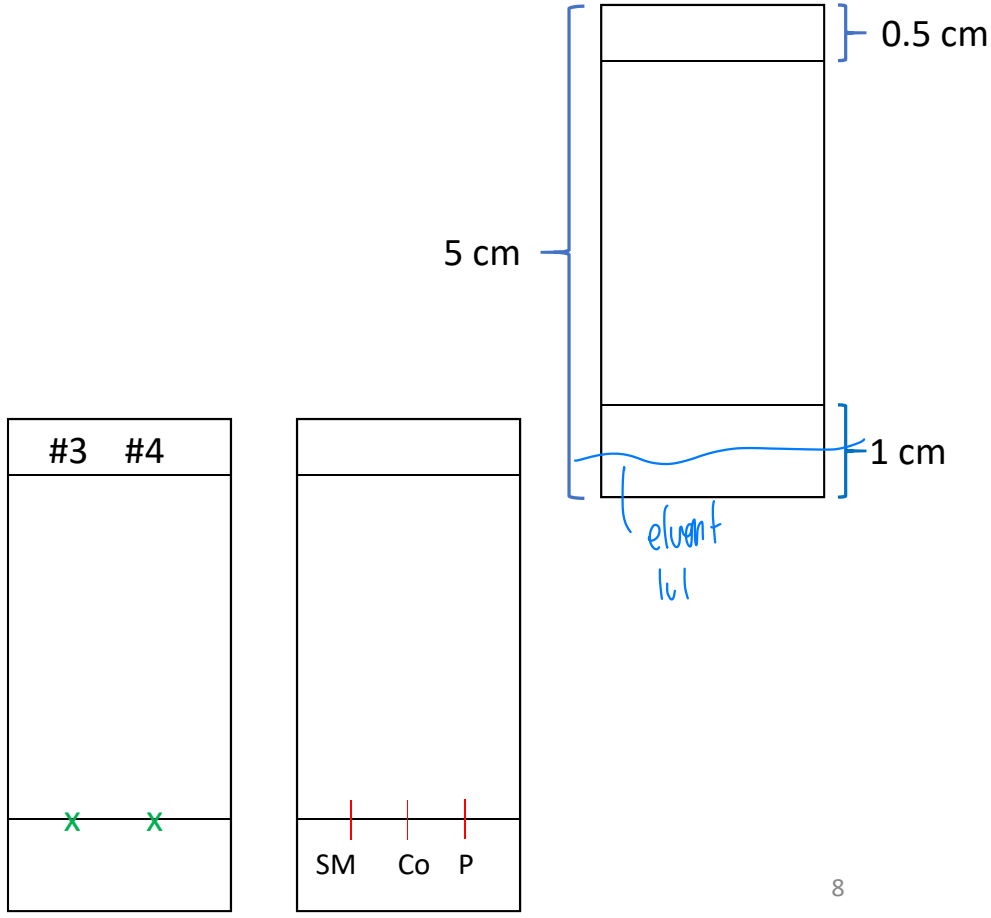
TLC: Chromaplate Drawing

- Use pencil to draw the starting line (1 cm to the lower edge) and finishing line (0.5 – 1 cm to the upper edge)

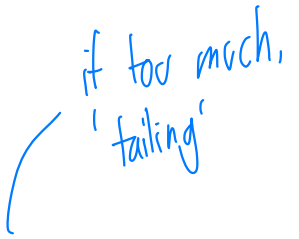
- CANNOT use pen!!!

When you become more experienced,
4-6 spots on one TLC plate is possible

- Mark 2-3 spots on the starting line, as positions to load your sample
- Label the spots either above the finishing line or below the starting line



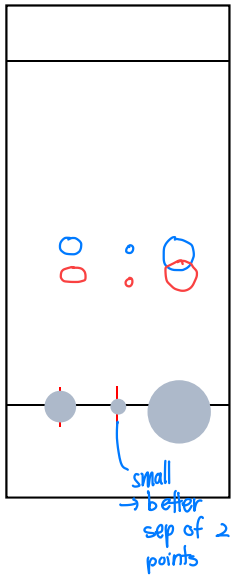
TLC: Sample Loading (1)



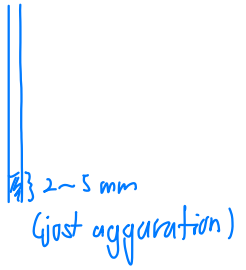
- Rule of thumb: the sample concentration for TLC is roughly 0.5-3.0%, e.g. dissolving 3 mg sample in 0.1 – 0.6 mL of solvent, depending on its solubility (may heat if solubility is poor at r.t.)
- Solvent should be reasonably volatile so that evaporation is fast
- Dip a capillary tube into the sample solution → observe the solution rise inside the capillary tube because of capillary action
- Gently touch the capillary tube to the marking on the chromaplate → observe a small circle of solution being delivered on the starting line → within a few seconds, the circle becomes invisible due to evaporation of solvent

TLC: Sample Loading (2)

Especially critical for the D-A reaction
Spot multiple times when necessary



- Is there sufficient amount of sample on the plate?
  - Check the plate under UV lamp
  - You should see a dark spot at your loading position
  - If the spot is faint, you need to apply the loading multiple times until the spot is dark enough → in between each loading, ensure the solvent fully evaporates, otherwise the circle will become bigger and bigger due to diffusion
- Circle is too big? Will that affect your TLC analysis?
  - A tip to control the volume of sample solution to be loaded: touch the capillary tube to a tissue paper/filter paper multiple times → each time the solution inside the capillary tube will decrease by a few mm → repeat until there is only around 2-5 mm of solution inside your capillary tube → load on to TLC plate
  - 0.5 µL solution in the capillary tube → circle diameter 3 mm on plate
- Capillary tube can be reused: cleaned by dipping into fresh solvent and draining by touching tissue paper, repeating 2-3 times

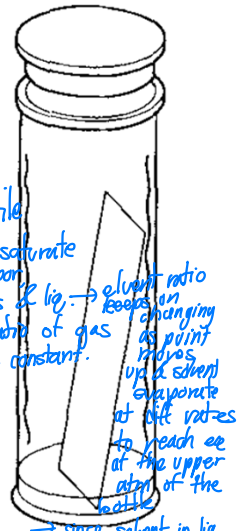


TLC: Development of Chromaplate

- Fill the glass bottle with 1~2 mL of the developing solvent (typically hexane + ethyl acetate) → the height of solvent is ~ 0.5 cm, which must be lower than the starting line of the TLC plate
- Place a piece of filter paper into the bottle, cap and wait for a few minutes → allow the atmosphere in the jar to be saturated with solvent vapour
- Put in the chromaplate using the tweezer and cap the bottle as soon as possible → the solvent will ascent by capillary action to the finishing line → uncap the bottle → take out the chromaplate → cap the bottle → dry the chromaplate

↳ Raoult's law & phase diagram.
2 volatile liq evaporate to give diff gaseous composition compared to liq. more volatile liq found more in gas.

↳ Hex is more volatile than EA, so if x saturate the chamber w vapor, x reach eq b/w gas & liq. → if eq, ratio of gas & liq is constant.



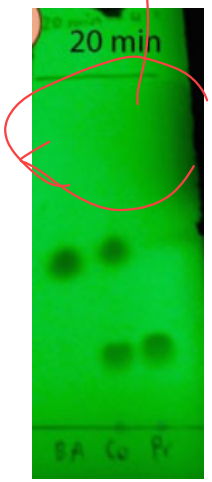
⇒ since solvent in liq ratio keeps on changing, result is NOT repeatable!

if go beyond finish line, can't identify solvent front

TLC: Observation of the Spots

ZnS is inorganic → × dissove in organic solvent.
↳ × abs UV. so give green Flu ↑

- If the components are coloured, they can be readily visualized, as long as the concentration is sufficiently high
- For compounds that are UV-active (e.g. containing conjugation system), fluorescent quenching effect is used to visualize the spot under UV lamp → circle the spot by pencil under UV lamp
- In case the compounds are not UV-active, various staining reagents are available:
  - Iodine vapour
  - $KMnO_4$ – $NaHCO_3$
  - p-anisaldehyde
  - Phosphomolybdic acid
  - Many others



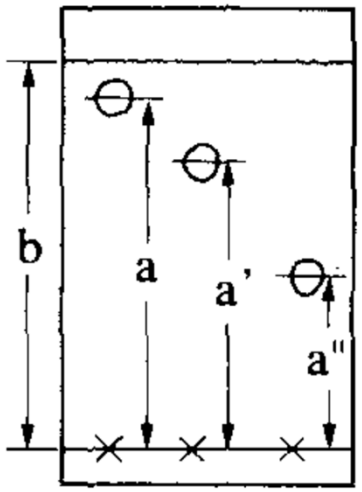
* shld NOT judge based on color.
need UV.
↳ e.g. Exp 2 Sonogashira has coloured SM & P.

↳ this is because desired product may be colorless and finished reactant is colored due to impurity.

TLC: R_f value

- $R_f = \dfrac{distance\ moved\ by\ substance}{distance\ moved\ by\ the\ solvent\ front}$

centre of circle



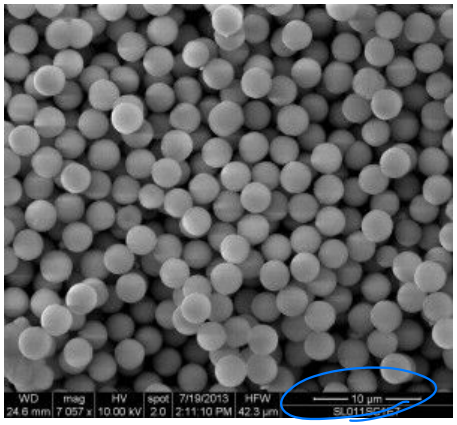
- Theoretically, R_f value is a constant for a given substance in a given solvent
- Practically, R_f value may have non-ideal reproducibility due to: solvent composition, degree of saturation of the tank atmosphere with solvent vapour, solvent "climbing up" the plate path not in a straight line
- R_f value is useful to differentiate components and serves as a guidance for the column chromatography separation

Part 2

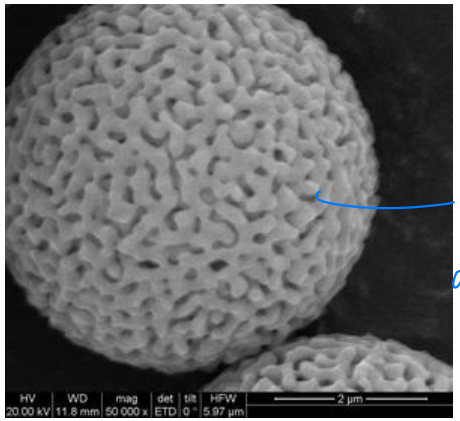
Column Chromatography

Column Chromatography

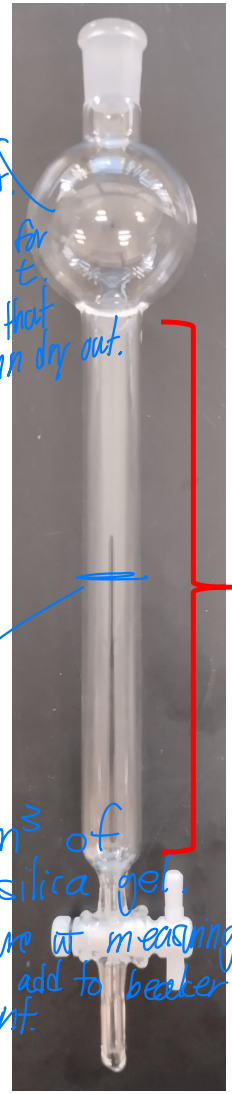
- A common separation technique used in synthesis
- Glass column: 10 – 90 cm long and 1-5 cm diameter
- With or without a bulb to hold more eluent
- Packing material: silica gel or alumina, ~50 μm in diameter



10 μm



lot of pores where adsorption · desorption · sep occur.



storage space for solvent to run column for longer t. but note that solvent can dry out.

바닥은 채움.

↳ 30 cm³ of dry silica gel measure at measuring cylinder & then add to beaker ∴ the solvent.

Diameter: ~ 2cm

Column Chromatography

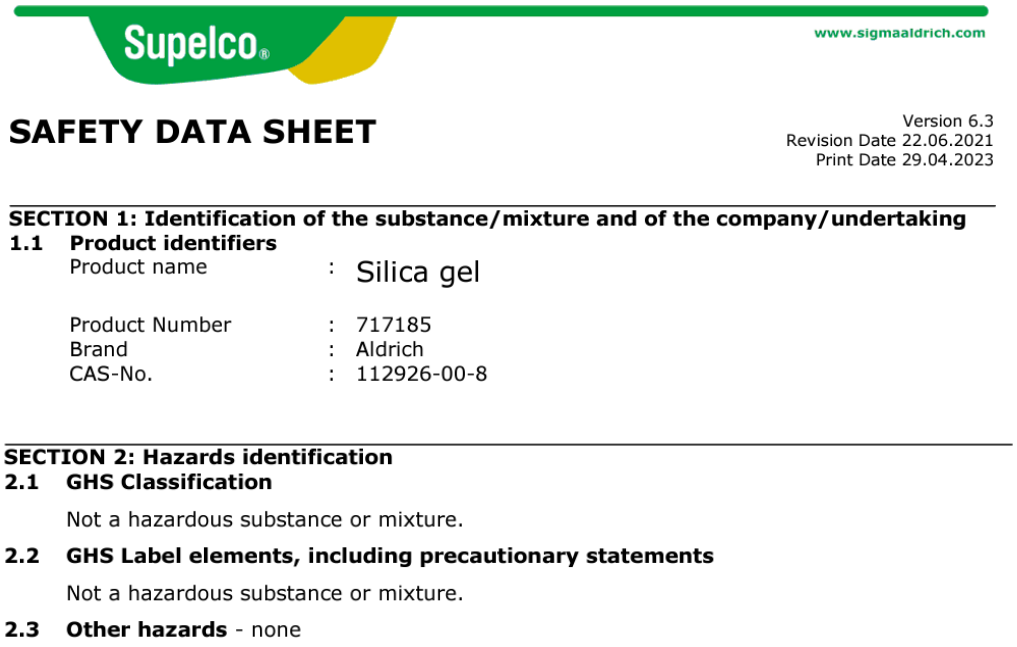
- Separation is via different adsorption of each component onto silica gel
- Generally, the more polar component (e.g. ketone) is adsorbed more strongly on the silica gel than the less polar component (e.g. hydrocarbon) → hydrocarbon is eluted faster than ketone → collected in separate test tubes and recovered separately → separation achieved
- Broadly, the ease of elution follows the following order: saturated hydrocarbons > alkenes, alkynes, arenes > esters, aldehydes, ketones > amines, alcohols, thiols > phenols, carboxylic acids

i.e. travels slower.

least attraction to silica gel

Column Chromatography: Silica Gel

- Silica gel is of very small particle size (50 µm)
- It is not as small as PM2.5, but if you breathe in silica gel particle, it may still trigger long-term damage to human body, due to lack of metabolism mechanism
- Keep the dry silica gel inside the fumehood
- Only transfer “wet” silica gel (e.g. slurry) between fumehoods



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SAFETY DATA SHEET

Version 6.3
Revision Date 22.06.2021
Print Date 29.04.2023

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifiers

Product name : Silica gel

Product Number : 717185
Brand : Aldrich
CAS-No. : 112926-00-8

SECTION 2: Hazards identification

2.1 GHS Classification

Not a hazardous substance or mixture.

2.2 GHS Label elements, including precautionary statements

Not a hazardous substance or mixture.

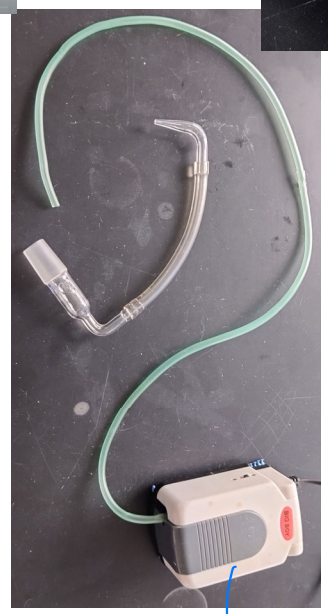
2.3 Other hazards - none

Column Chromatography: Column Packing

- Column chromatography with coloured eluant(s)
  - https://www.youtube.com/watch?v=KGTZ3XBEfyc
  - Always inside the fumehood
  - When transporting silica gel from common fumehood to your fumehood → in slurry form
- Column chromatography with colourless eluant(s)
  - https://www.youtube.com/watch?v=u8yfYan6thA

desired product
$R_f \approx 0.3$ 이 되도록
solvent ratio 조정.



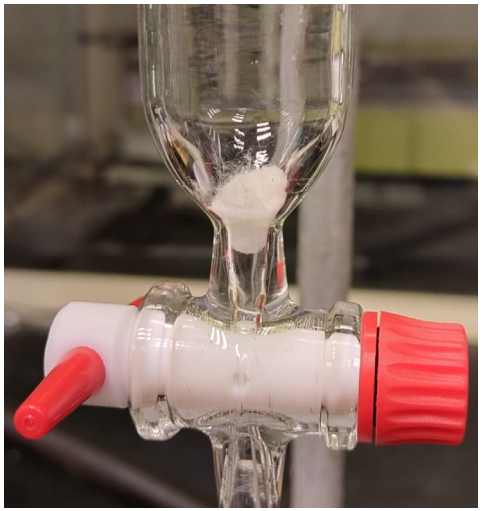


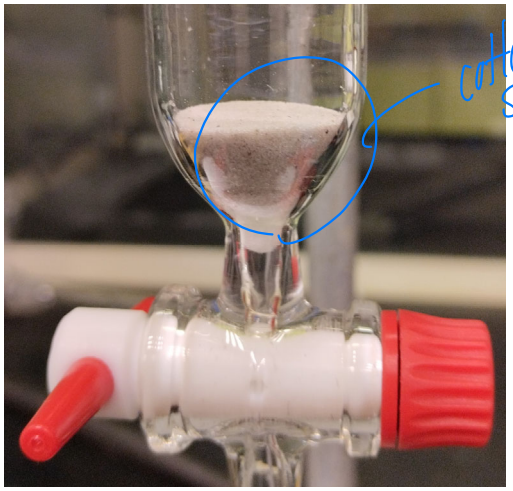


Column Chromatography: Column Packing (1)

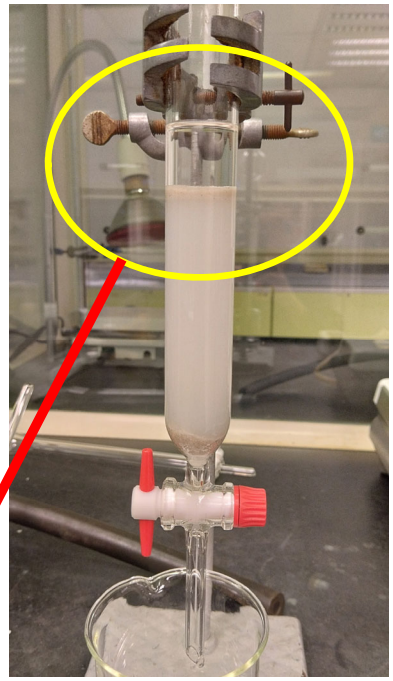
- The glass column must be vertical (not slanted)
- The cotton wool cannot be too tight (flow rate will be slow) or too loose (silica gel pass through)
- Silica gel column shall not have any cracks or bubbles → the level of liquid shall NOT fall below the level of silica gel
- V(silica gel) ~ 30 mL → 9-12 cm height inside column
  - Use a common measuring cylinder + solid funnel to measure silica gel in the common fumehood. Do not take away the common measuring cylinder/funnel.
  - Bring your own beaker, with the eluent inside → so that silica is in slurry form
- Reuse the eluent from packing, since it is clean (free from sample/impurity)

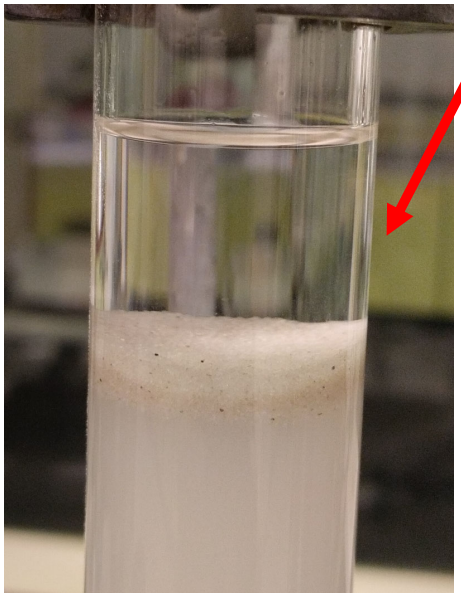
The volume has been pre-measured for you

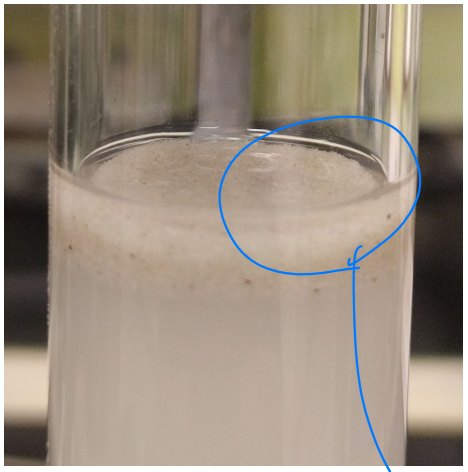




cotton 대신에 sand 넣기.







Drain slowly until eluent just covers the silica / sand layer

1 cm 높이 될 때까지만 air pump로 solvent 빼내기.
남은 양은 자연적으로 drain

Column Chromatography: Column Packing (2)

- Sand layer on top of the silica gel column is optional
  - Protect the silica gel column from disturbance, but if the disturbance is too much, protection can still fail
  - Sand layer might be added either before or after loading the sample
  - Separation occurs on silica gel instead of sand layer, so a broken glass dropper tip stuck in the sand layer will not affect your separation
- Air pump is optional
  - When the flow rate is too slow, using the pump help to speed up the entire separation progress
  - REMEMBER to remove the air pump when solvent level is low, especially when loading the sample
  - Always ensure the tubing is not bent



Column Chromatography: Sample Loading

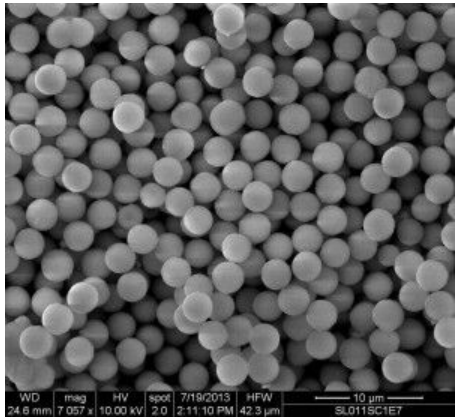
- Dissolve the solid sample in an appropriate solvent (ideally the same as the eluent)
- The volume of solvent shall be as minimum as possible, otherwise band overlap might occur
- Rinse of the sample vial / column wall shall be minimum too
- Add as gently as possible, to minimize disturbance to the silica gel
- Before the next addition / rinse, ensure the sample is adsorbed onto silica gel by careful draining the eluent (control the tap carefully)

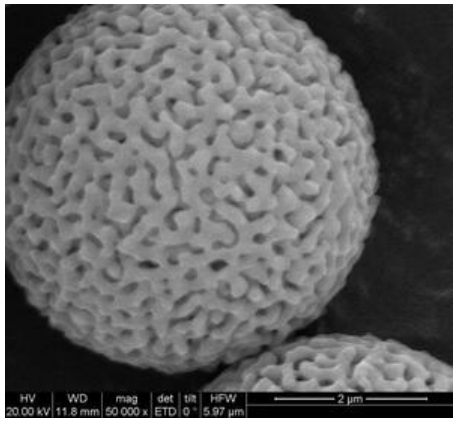
Column Chromatography: Sample Loading

- Use a glass dropper to add this concentrated solution to the packed silica gel column
  - Do NOT drop the solution directly onto the silica gel → “drill” a hole and not using the entire column effectively
  - Add the solution along the inner wall, so that the solution slowly flows down until it touches the silica gel column and get adsorbed
  - Add the solution in circular manner, so that the entire surface of silica gel is used to adsorb the sample
  - Diffusion starts once your sample is loaded → if you would like to go for lunch, should be before loading your sample

Column Chromatography: R_f and separation

- R_f value from TLC analysis can give us a clue on how well the column chromatography separation is going to be
- Column volume (CV): volume of mobile phase needed to fill all the void volume in a packed column → all the "empty" space after the column is properly packed, e.g. the gaps between the spherical particles and the pores of each particle
- CV number: the number of CVs needed to elute a component





Column Chromatography: R_f and separation

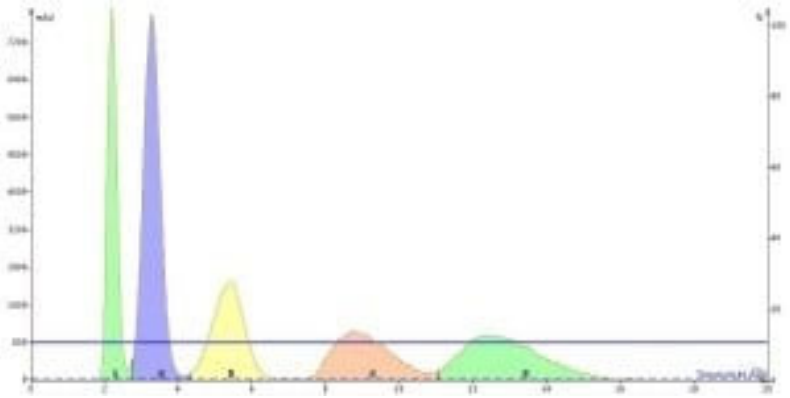
- Using the same stationary and mobile phase in TLC and column chromatography: CV number = $1/R_f$
- How to estimate Column Volume (CV)?
  - 100 mL eluent is used to pack the column → after lowering the eluent level to just the same as silica gel level, the left-over eluent is 80 mL → 20 mL is in the "wet" silica gel column → CV = 20 mL
- Let's consider two hypothetical components A and B: R_f(A)= 0.30, R_f(B) = 0.20
  - CV number (A) = 1/0.30 = 3.33 → 67 mL to elute A
  - CV number (B) = 1/0.20 = 5.00 → 100 mL to elute B
  - If you collect 10 mL eluent in each test tube, A will be in Tube #? How about B?

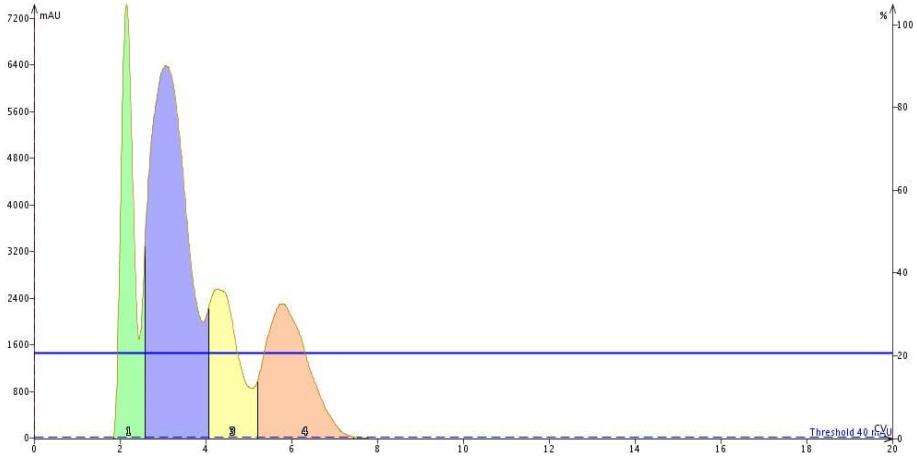
Column Chromatography: R_f and separation

- CV number (A) = 1/0.30 = 3.33 → 67 mL to elute A
- CV number (B) = 1/0.20 = 5.00 → 100 mL to elute B
- The values obtained is often not accurate
  - R_f value has a relatively large error → distance is only a few cm
  - Two eluents may not be of exactly identical ratio, particularly when one component is of very small percentage (e.g. 1%, 0.5%, etc.)
  - CV is just an estimation
- But the calculation still provide clue on how much we shall collect in each test tube: likely to collect ~10 mL in each test tube, rather than 20-30 mL in each test tube
  - If R_f = 0.50/0.40 → 40 mL / 50mL → clearly you cannot collect 10 mL in each test tube → 2-4 mL in each test tube is safer → better change the eluent for a smaller R_f

Column Chromatography: R_f and separation

- The calculation does not consider diffusion, e.g. assuming all of component A is eluted at exactly the same time within one drop of eluent, but that is not the reality
- In reality, diffusion always occurs and cannot be prevented
- When too much sample is used, it may “overload” the column and result in component elution overlap





Column Chromatography: Rule of thumb

- Ratios of those solvents should be adjusted to get the targeted compound to an R_f between 0.1 and 0.4 [0.35 according to]
- Choose the column diameter according to the sample mass to be separated
- Pack the column to the height of 9 – 12 cm

Rapid Chromatographic Technique for Preparative Separations with Moderate Resolution

W. Clark Still,* Michael Kahn, and Abhijit Mitra

Table I

column diameter, mm	vol of eluant,[a] mL	sample: typical loading (mg) $\Delta R_f \geq 0.2$	sample: typical loading (mg) $\Delta R_f \geq 0.1$	typical fraction size, mL
10	100	100	40	5
20	200	400	160	10
30	400	900	360	20
40	600	1600	600	30
50	1000	2500	1000	50

[a] Typical volume of eluant required for packing and elution.

J. Org. Chem., Vol. 43, No. 14, 1978 **2923**

nent on analytical TLC to an R_f of 0.35.[9] If several compounds are to be separated which run very close on TLC, adjust the solvent to put the midpoint between the components at R_f = 0.35. If the compounds are widely separated, adjust the R_f of the less mobile component to 0.35. Having chosen the solvent, a column of the appropriate diameter

Column Chromatography: eluent collection and evaluation

- Guideline is for experienced user → beginners may need to increase(?)/decrease(?) the collection volume in each test tube
- How to know the volume of X mL in a test tube?
- How to know #? test tube contains product or not and which product?

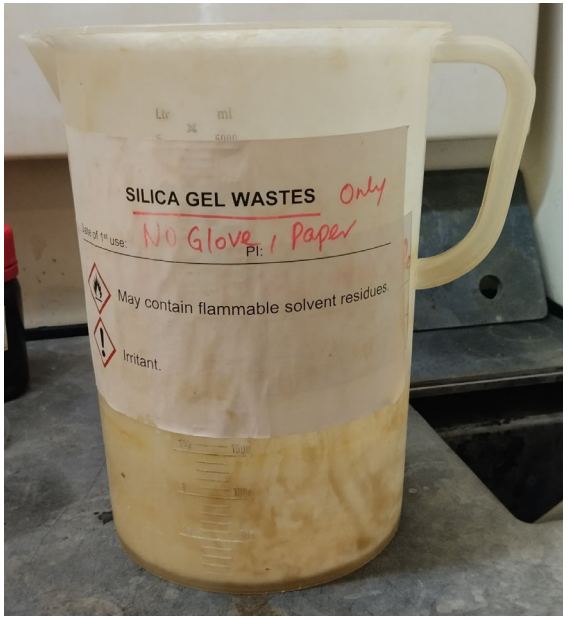
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10	100	100	40	5
20	200	400	160	10
30	400	900	360	20
40	600	1600	600	30
50	1000	2500	1000	50

[a] Typical volume of eluant required for packing and elution.

Clean up

- After separation is confirmed and product obtained → push out all solvent from the column by use of the air pump (30-40 min) until the silica gel is fully dry
- Bring the column to the waste solvent fumehood and pour the dry silica gel into the “Silica Gel Waste” container
- If the silica gel is properly dried, the silica gel shall have no problem going into the container; otherwise, dry the silica gel further by air pump



Clean up

- The remaining eluent (from rotavap and column drying) shall be disposed into the waste solvent bottle
- Chlorinated?
- Non-chlorinated?



Clean up

- To clean the glassware, an appropriate solvent is used to dissolve the stains
- The stain may not be organic, so organic solvent may not always be the choice
- Acetone is a common choice (but not necessarily the only choice and the correct choice)
- The wash shall be collected and disposed properly

